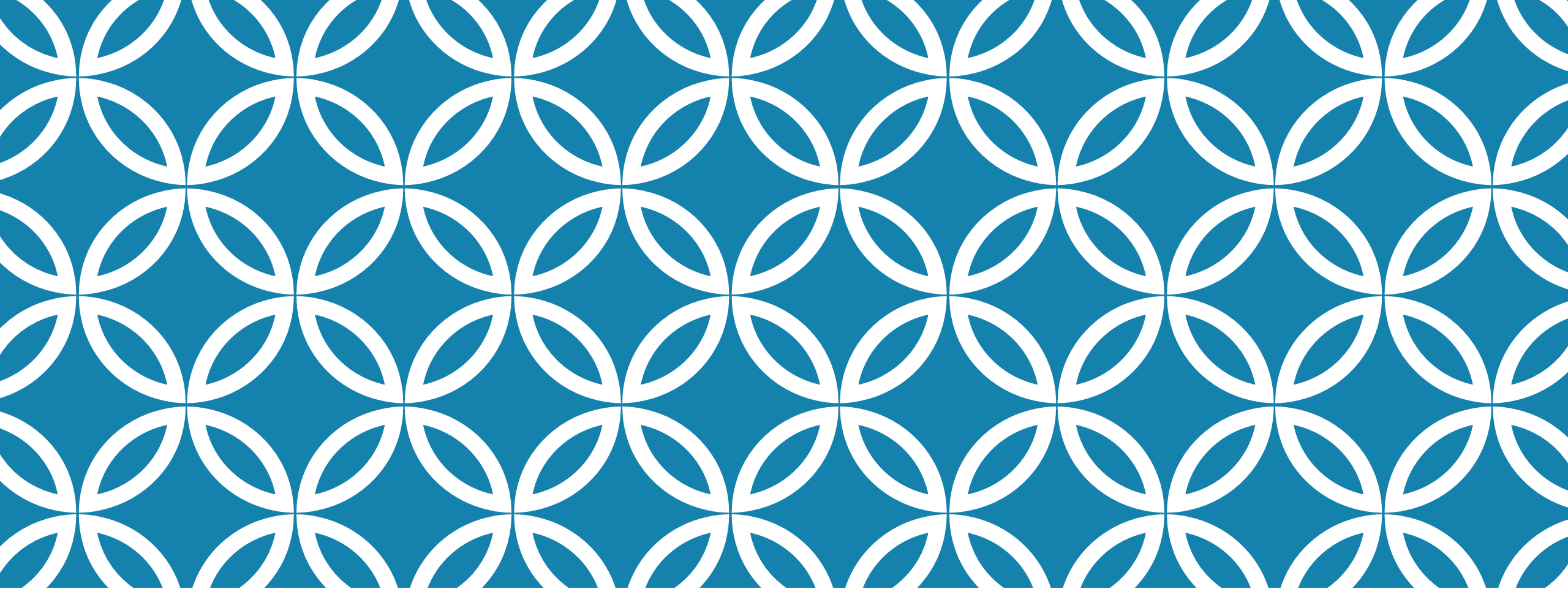




FUZZY SET THEORY

-Dr. Zainab Mirza

CONTENTS

- Fuzzy sets
- Fuzzy properties
- Fuzzy operations
- Fuzzy relations
- Fuzzy composition
- Fuzzy inference
- Defuzzification methods

CLASSICAL SET

- A set is a collection of distinct elements without ordering and without repetition.
- Defined by crisp boundaries
- No uncertainty about the set boundaries
- Fuzzy set is defined by its ambiguous boundaries

Why?

- Real world data is not precise
- Fuzziness/vagueness/ inexactness

FUZZY SET THEORY

Examples

Satish is a **rich** man

That blue car is **expensive**

No well-defined demarcations between

- rich and poor
- very and little
- expensive and cheap
- old and young
- weak and strong

The classical set theory is not equipped to handle such vagueness as it does not allow an element to be a partial member

Fuzzy set theory is a generalization of the classical set theory

FUZZY SET THEORY

Fuzzy set theory was proposed by Zadeh in 1965

A fuzzy set is defined as a set of ordered pairs where the first element is the element and the second element is membership of that element

$$A = \left\{ (x, \underbrace{\mu_A(x)}_{\substack{\text{membership} \\ \text{function}}}) \mid x \in U \right\}$$

$$\mu_A : U \rightarrow [0, 1]$$

U : universe of discourse.

REPRESENTATION

$$A = \left\{ \begin{array}{cccc} (1, 0.1) & (2, 0.3) & (3, 0.8) & (4, 1) \\ (5, 0.9) & (6, 0.5) & (7, 0.2) & (8, 0.1) \end{array} \right\}$$

Alternative Representation:

$$A = 0.1/1 + 0.3/2 + 0.8/3 + 1.0/4 + 0.9/5 + 0.5/6 + 0.2/7 + 0.1/8$$

Properties of fuzzy sets

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup A = A$$

$$A \cap A = A$$

$$A \cup X = X$$

$$A \cap X = A$$

$$A \cup \emptyset = A$$

$$A \cap \emptyset = \emptyset$$

If $A \subseteq B \subseteq C$, then $A \subseteq C$

$$A'' = A$$

Properties of fuzzy sets

De Morgan's Law also holds:

$$(A \cap B)' = A' \cup B'$$

$$(A \cup B)' = A' \cap B'$$

FUZZY SET OPERATIONS

Union

$$\begin{aligned}\mu_{A \cup B}(x) &= \mu_A(x) \cup \mu_B(x) \\ &= \max(\mu_A(x), \mu_B(x))\end{aligned}$$

Intersection

$$\begin{aligned}\mu_{A \cap B}(x) &= \mu_A(x) \cap \mu_B(x) \\ &= \min(\mu_A(x), \mu_B(x))\end{aligned}$$

Compliment

$$\mu_{A'}(x) = 1 - \mu_A(x)$$

FUZZY SET OPERATIONS: EXAMPLE

$$\bar{A} = \{0.1/1 + 0.2/2 + 0.3/3\}$$

$$\bar{B} = \{0.6/1 + 0.5/2 + 0.4/3 + 0.5/4\}$$

$$\bar{A} \cup \bar{B} = \{0.6/1 + 0.5/2 + 0.4/3 + 0.5/4\}$$

$$\bar{A} \cap \bar{B} = \{0.1/1 + 0.2/2 + 0.3/3 + 0/4\}$$

$$\bar{A}' = \{0.9/1 + 0.8/2 + 0.7/3 + 1/4\}$$

$$\bar{B}' = \{0./1 + 0.5/2 + 0.6/3 + 0.5/4\}$$

FUZZY SET OPERATIONS

Comparison

Is $\bar{A} = \bar{B}$?

$\bar{A} = \bar{B}$ if $\mu_{\bar{A}}(x) = \mu_{\bar{B}}(x) \quad \forall x \in X$

Containment

Is $\bar{A} \subset \bar{B}$?

$\bar{A}$ is in $\bar{B}$ if $\mu_{\bar{A}}(x) < \mu_{\bar{B}}(x) \quad \forall x \in X$

FUZZY SET OPERATIONS:EXAMPLE

$$\bar{A} = \{0.1/1 + 0.2/2 + 0.3/3\}$$

$$\bar{B} = \{0.6/1 + 0.5/2 + 0.4/3 + 0.5/4\}$$

4) The 2 sets are not equal

5) $\bar{A}$ is contained in $\bar{B}$

FUZZY MEMBERSHIP FUNCTION

- Membership function $\mu_A(x)$ defines degree of membership of x in A
- It assumes value in the range $[0,1]$

E.g. $\mu_A(x) = 0.6$

i.e. x belongs to Set A by 0.6 value

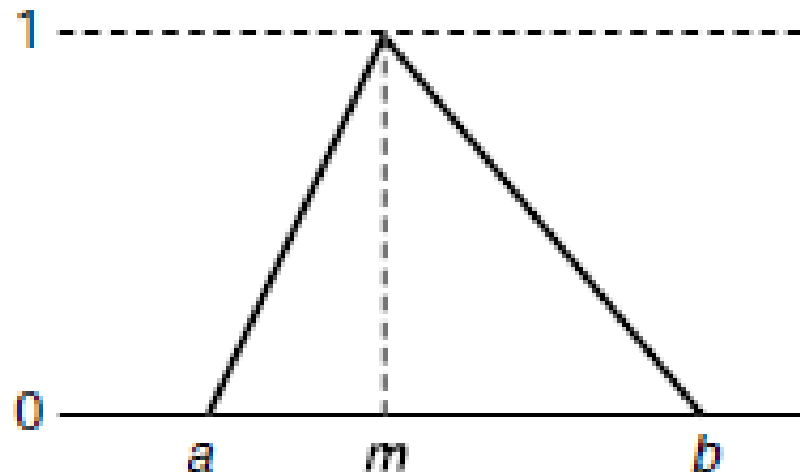
FUZZY MEMBERSHIP FUNCTION

The most concise way to define a MF is to express it as a mathematical formula

$$\mu(x) = \begin{cases} 0, & \text{if } x \leq a \\ \frac{x-a}{m-a}, & \text{if } a \leq x \leq m \\ \frac{b-x}{b-m}, & \text{if } m \leq x \leq b \\ 0, & \text{if } x \geq b \end{cases}$$

SOME POPULAR FUZZY MEMBERSHIP FUNCTIONS

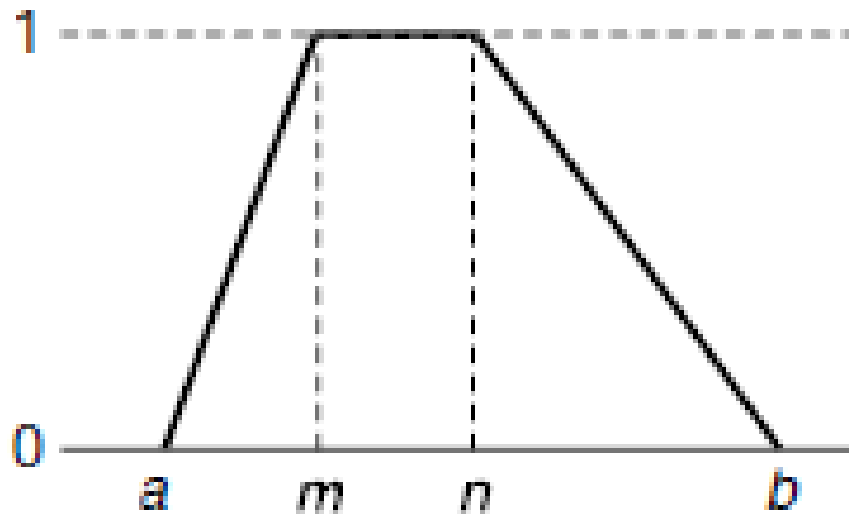
Triangular function- the most frequently used membership function



$$\mu(x) = \begin{cases} 0, & \text{if } x \leq a \\ \frac{x-a}{m-a}, & \text{if } a \leq x \leq m \\ \frac{b-x}{b-m}, & \text{if } m \leq x \leq b \\ 0, & \text{if } x \geq b \end{cases}$$

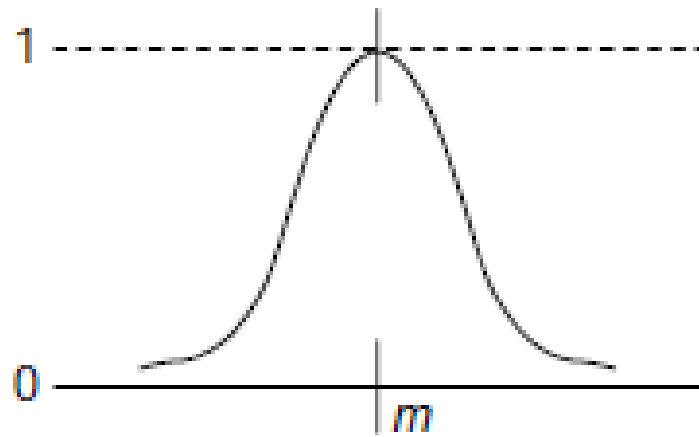
SOME POPULAR FUZZY MEMBERSHIP FUNCTIONS

Trapezoidal function

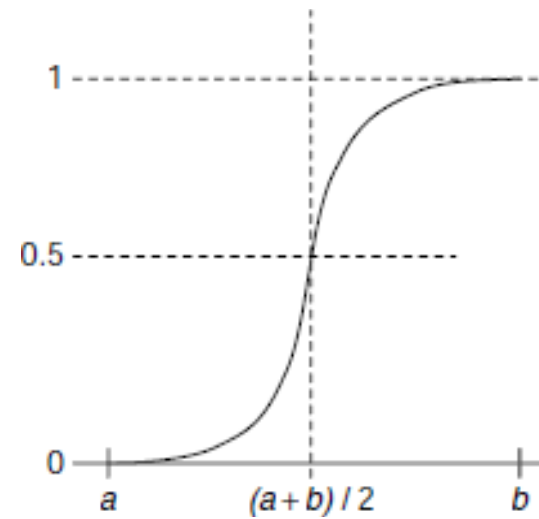


$$\mu(x) = \begin{cases} 0, & \text{if } x \leq a \\ \frac{x-a}{m-a}, & \text{if } a \leq x \leq m \\ 1, & \text{if } m \leq x \leq n \\ \frac{b-x}{b-n}, & \text{if } n \leq x \leq b \\ 0, & \text{if } x \geq b \end{cases}$$

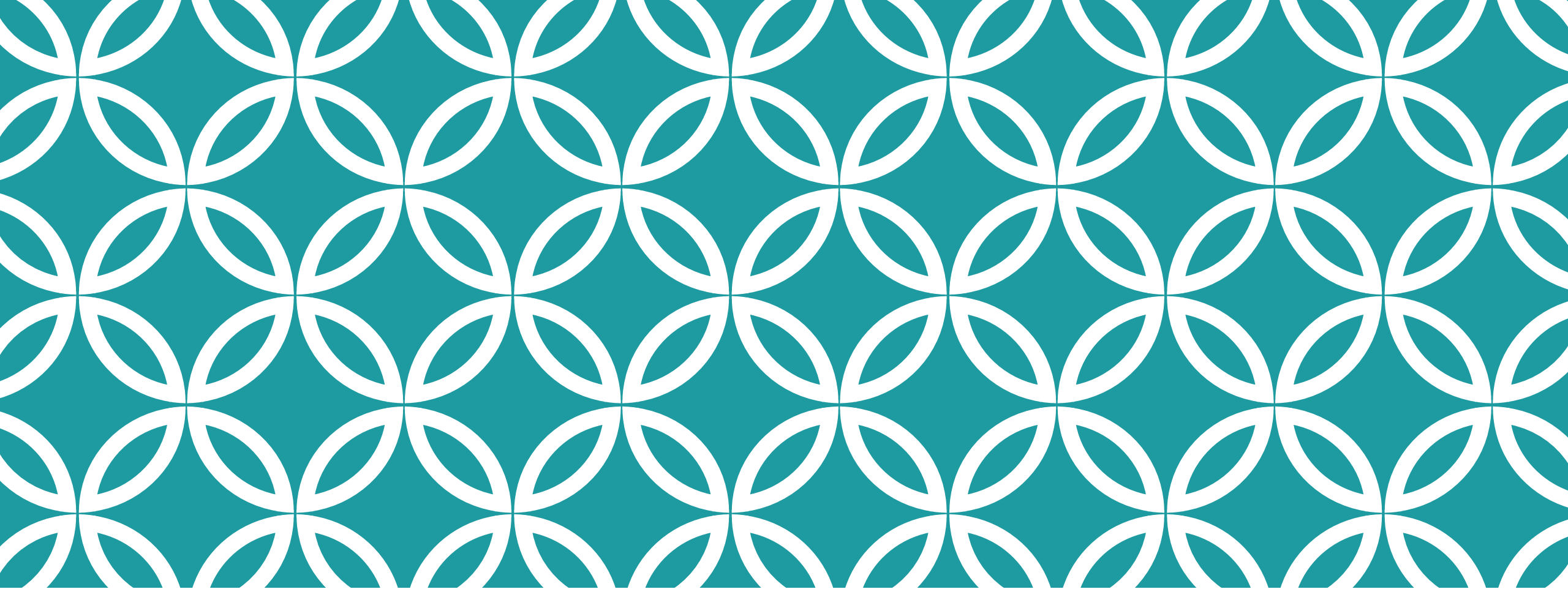
SOME POPULAR FUZZY MEMBERSHIP FUNCTIONS



Gaussian function



S-function

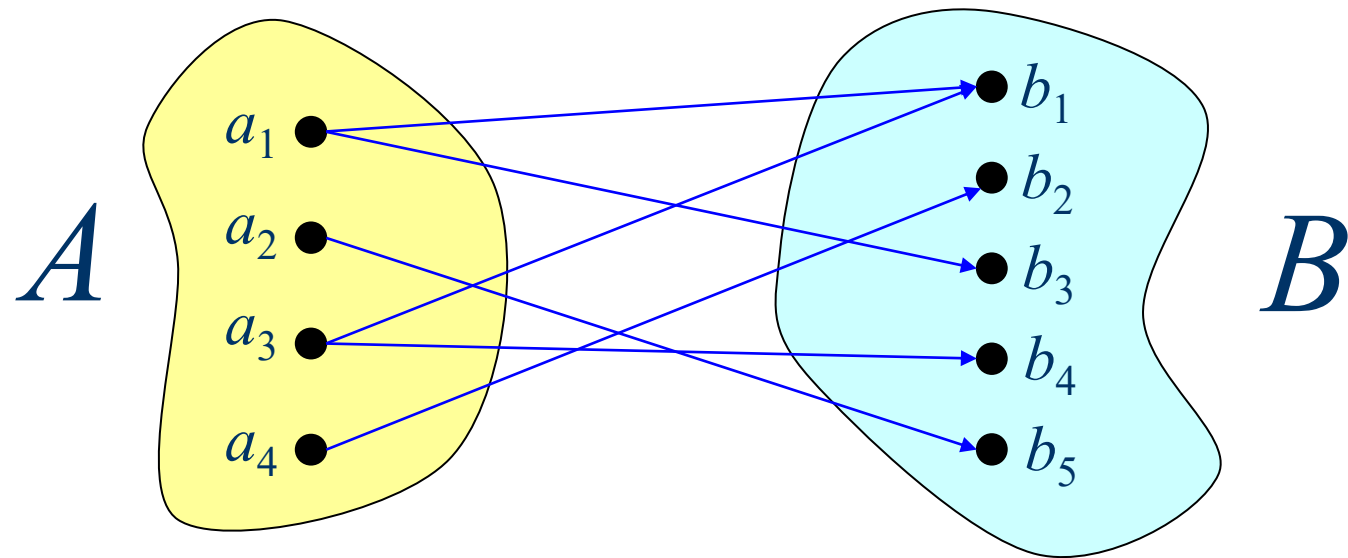


FUZZY RELATIONS

THE REAL-LIFE RELATION

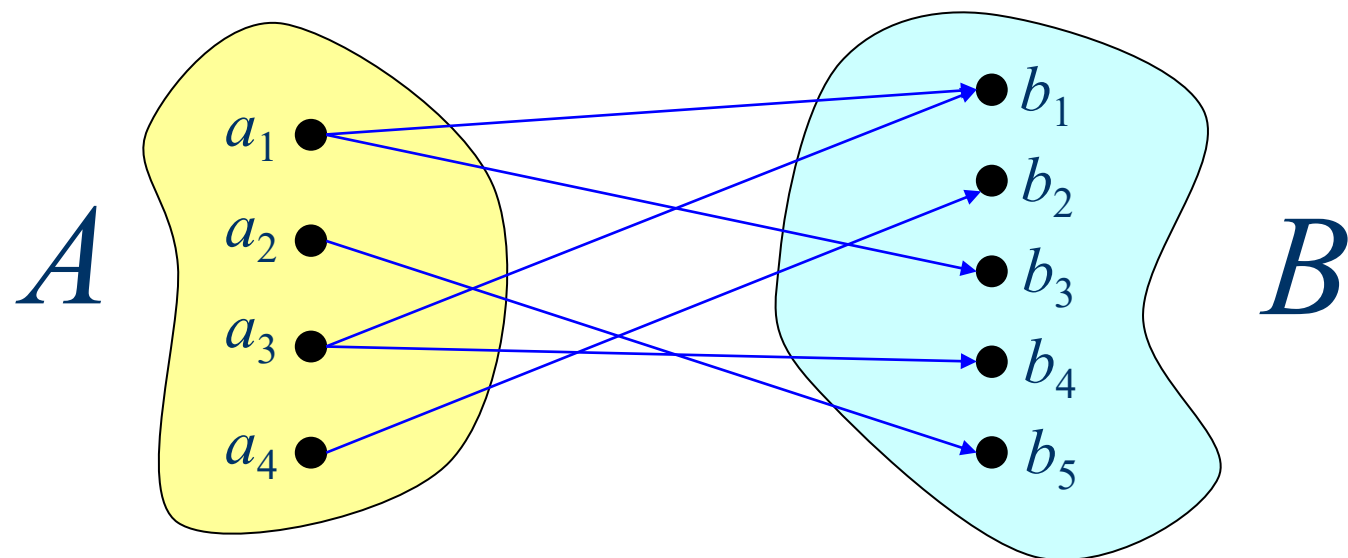
- x is close to y
 - x and y are numbers
- x depends on y
 - x and y are events
- x and y look alike
 - x and y are persons or objects

CRISP RELATION (R)



$$\underline{R} \subseteq A \times B$$

CRISP RELATION (R)



$$M_R = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

$$R \subseteq A \times B$$

Crisp Relations

Example:

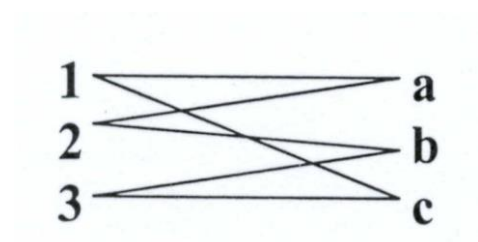
If $X = \{1,2,3\}$

$Y = \{a,b,c\}$

$R = \{ (1\ a), (1\ c), (2\ a), (2\ b), (3\ b), (3\ c) \}$

$$R = \begin{matrix} & \begin{matrix} a & b & c \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \end{matrix}$$

Using a diagram to represent the relation



FUZZY RELATIONS

Triples showing connection between two sets:

$(a,b,\#)$: a is related to b with degree $\#$

Fuzzy relations are set themselves

Fuzzy relations can be expressed as matrices

...

FUZZY RELATIONS MATRICES

Example: relation between set of diseases and set of symptoms

$R_1(x, y)$	Running nose	High temp	shivering
Typhoid	0.1	0.9	0.8
Viral fever	0.2	0.9	0.7
Common cold	0.9	0.4	0.6

FUZZY RELATIONS

Let $A = B = C = \{0, 1, 2, 3\}$ and the relations R , S , and T defined as follows :

$$R \subseteq A \times B, R = \{(a, b) \mid a + b \text{ is an even number}\}$$

$$S \subseteq A \times B, S = \{(a, b) \mid b = (a + 2) \text{ MOD } 3\}$$

$$T \subseteq B \times C, T = \{(b, c) \mid |b - c| = 1\}$$

These relations can be explicitly written as

$$R = \{(0, 0), (0, 2), (1, 1), (1, 3), (2, 0), (2, 2), (3, 1), (3, 3)\}$$

$$S = \{(0, 2), (1, 0), (2, 1), (3, 2)\}, \text{ and}$$

$$T = \{(0, 1), (1, 0), (1, 2), (2, 1), (2, 3), (3, 2)\}.$$

FUZZY RELATIONS

$$R = \{(0, 0), (0, 2), (1, 1), (1, 3), (2, 0), (2, 2), (3, 1), (3, 3)\}$$

$$S = \{(0, 2), (1, 0), (2, 1), (3, 2)\}, \text{ and}$$

$$T = \{(0, 1), (1, 0), (1, 2), (2, 1), (2, 3), (3, 2)\}.$$

$$T_R = \begin{array}{c} \begin{array}{cccc} & 0 & 1 & 2 & 3 \end{array} \\ \begin{array}{c} 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \end{array} \quad T_S = \begin{array}{c} \begin{array}{cccc} & 0 & 1 & 2 & 3 \end{array} \\ \begin{array}{c} 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{array} \quad T_T = \begin{array}{c} \begin{array}{cccc} & 0 & 1 & 2 & 3 \end{array} \\ \begin{array}{c} 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

OPERATIONS ON FUZZY RELATIONS

Union $(R \cup S)(x, y) = \max \{R(x, y), S(x, y)\}$

Intersection $(R \cap S)(x, y) = \min \{R(x, y), S(x, y)\}$

Complementation $R'(x, y) = 1 - R(x, y)$

OPERATIONS ON FUZZY RELATIONS

$$T_R = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \end{matrix} \quad T_S = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad T_{\bar{r}} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$T_{R \cup S} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} \quad T_{R \cap S} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad T_{R'} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

COMPOSITION

- Let R be a relation that relates, or maps, elements from universe X to universe Y , and let S be a relation that relates, or maps, elements from universe Y to universe Z .
- Can we find a relation, T , that relates the same elements in universe X that R contains to the same elements in universe Z that S contains?
- We can find such a relation using an operation known as *composition*.

	R	a	b	c	d	Y		S	α	β	γ	Z
	1	0.1	0.2	0.0	1.0			a	0.9	0.0	0.3	
	2	0.3	0.3	0.0	0.2			b	0.2	1.0	0.8	
	3	0.8	0.9	1.0	0.4			c	0.8	0.0	0.7	
								d	0.4	0.2	0.3	

$T?$

COMPOSITION

There are two common forms of the composition operation:

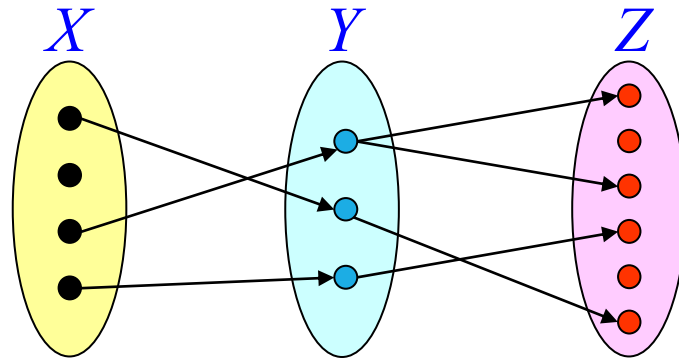
max-min composition

$$T = R \circ S,$$
$$\mu_T(x, z) = \bigvee_{y \in Y} (\mu_R(x, y) \wedge \mu_S(y, z)).$$

max-product composition.

$$T = R \circ S,$$
$$\mu_T(x, z) = \bigvee_{y \in Y} (\mu_R(x, y) \bullet \mu_S(y, z)).$$

MAX-MIN COMPOSITION



R : fuzzy relation defined on X and Y .

S : fuzzy relation defined on Y and Z .

$R \circ S$: the composition of R and S .

A fuzzy relation defined on X and Z .

$$\begin{aligned}\mu_{R \circ S}(x, z) &= \max_y \min(\mu_R(x, y), \mu_S(y, z)) \\ &= \vee_y (\mu_R(x, y) \wedge \mu_S(y, z))\end{aligned}$$

EXAMPLE

$$\mu_{S \circ R}(x, y) = \max_v \min(\mu_R(x, v), \mu_S(v, y))$$

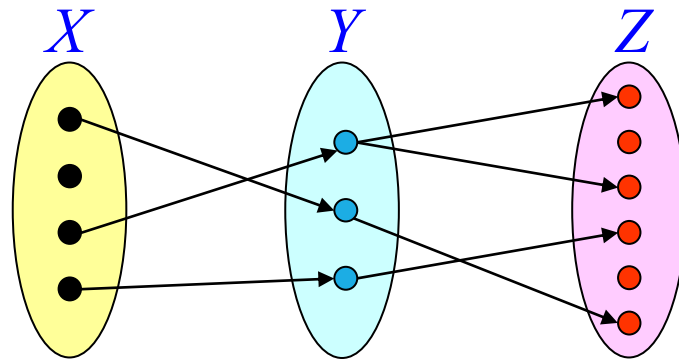
R	a	b	c	d
1	0.1	0.2	0.0	1.0
2	0.3	0.3	0.0	0.2
3	0.8	0.9	1.0	0.4

	0.1	0.2	0.0	1.0
min	0.9	0.2	0.8	0.4
max	0.1	0.2	0.0	0.4

S	α	β	γ
a	0.9	0.0	0.3
b	0.2	1.0	0.8
c	0.8	0.0	0.7
d	0.4	0.2	0.3

$R \circ S$	α	β	γ
1	0.4	0.2	0.3
2	0.3	0.3	0.3
3	0.8	0.9	0.8

MAX-PRODUCT COMPOSITION



R : fuzzy relation defined on X and Y .

S : fuzzy relation defined on Y and Z .

$R \circ S$: the composition of R and S .

A fuzzy relation defined on X and Z .

$$\mu_{R \circ S}(x, y) = \max_v \left(\mu_R(x, v) \mu_S(v, y) \right)$$

EXAMPLE

$$\mu_{R \circ S}(x, y) = \max_v (\mu_R(x, v) \mu_S(v, y))$$

R	a	b	c	d
1	0.1	0.2	0.0	1.0
2	0.3	0.3	0.0	0.2
3	0.8	0.9	1.0	0.4

0.1 0.2 0.0 1.0

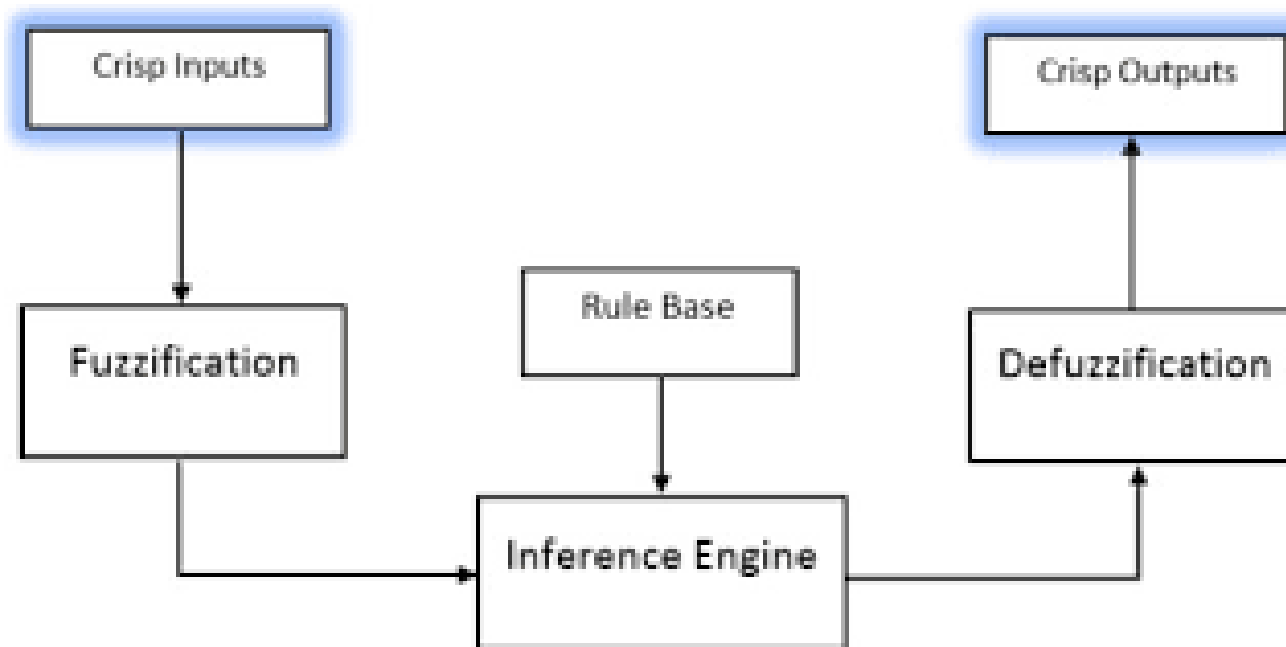
Product 0.9 0.2 0.8 0.4

max .09 .04 0.0 0.4

S	α	β	γ
a	0.9	0.0	0.3
b	0.2	1.0	0.8
c	0.8	0.0	0.7
d	0.4	0.2	0.3

$R \circ S$	α	β	γ
1	0.4	0.2	0.3
2	0.27	0.3	0.24
3	0.8	0.9	0.7

FUZZY INFERENCE PROCESS



FUZZY INFERENCE PROCESS

1. Fuzzification – input variables are assigned degrees of membership in various classes
2. Rule evaluation – inputs are applied to a set of if-then control rules
E.g. if temperature is very hot then set fan speed to very high
3. Defuzzification – fuzzy outputs are combined into discrete values needed to drive the actual mechanism

FUZZY CONTROLLER

Design a fuzzy controller to determine the wash time of a domestic washing machine. Assume that input is dirt, grease on clothes. Use 3 descriptors for input variables and 5 descriptors for output variable. Device a set of rules for control action & defuzzification. Clearly indicate that if the clothes are soiled to a larger degree the wash time required will be more.

FUZZY CONTROLLER

Step 1: Identification of i/p and o/p using linguistic variables

Step 2: Assign membership function to the variables

Step 3: Build a rule base

Step 4: Generate a crisp controlled output

FUZZY CONTROLLER

Step 1: Identification of i/p and o/p using linguistic variables

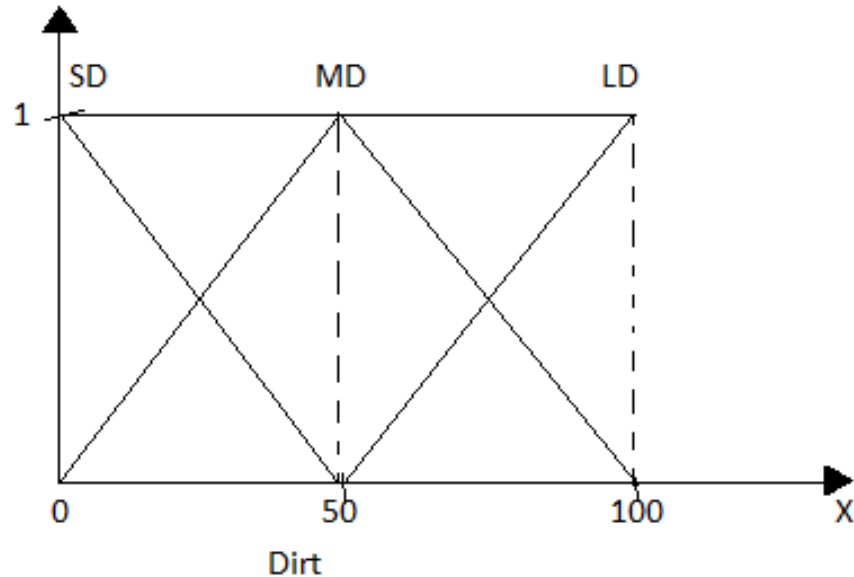
Input 1: Dirt $\Rightarrow$ { Small D, Medium D, Large D }

Input 2: Grease $\Rightarrow$ {Small G, Medium G, Large G }

Output: Wash time $\Rightarrow$ {Very Short, Short, Medium, Long, Very Long}

FUZZY CONTROLLER

Step 2: Assign membership function to the variables

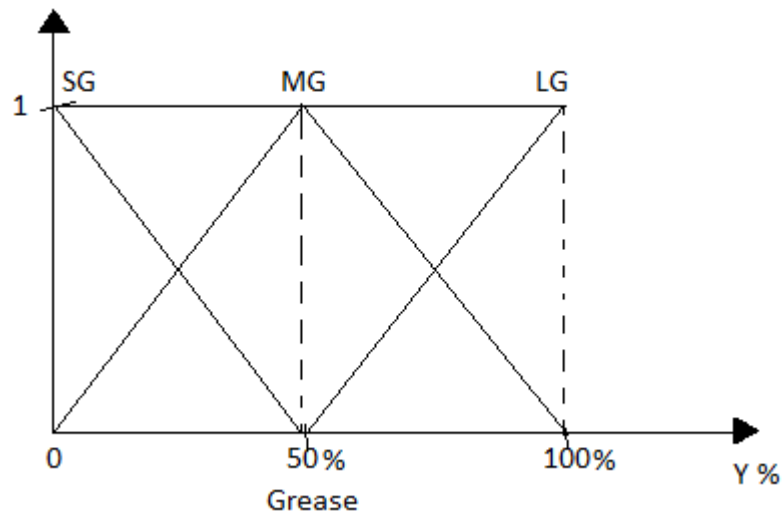


$$\mu_{\text{dirt}}(x) = \mu_{\text{SD}}(x) = (50-x)/50, \quad 0 \leq x \leq 50$$

$$\begin{aligned} &= \mu_{\text{MD}}(x) = (x-0)/50, \quad 0 \leq x \leq 50 \\ &= (100-x)/50, \quad 50 < x \leq 100 \end{aligned}$$

$$= \mu_{\text{LD}}(x) = (x-50)/50, \quad 50 \leq x \leq 100$$

FUZZY CONTROLLER

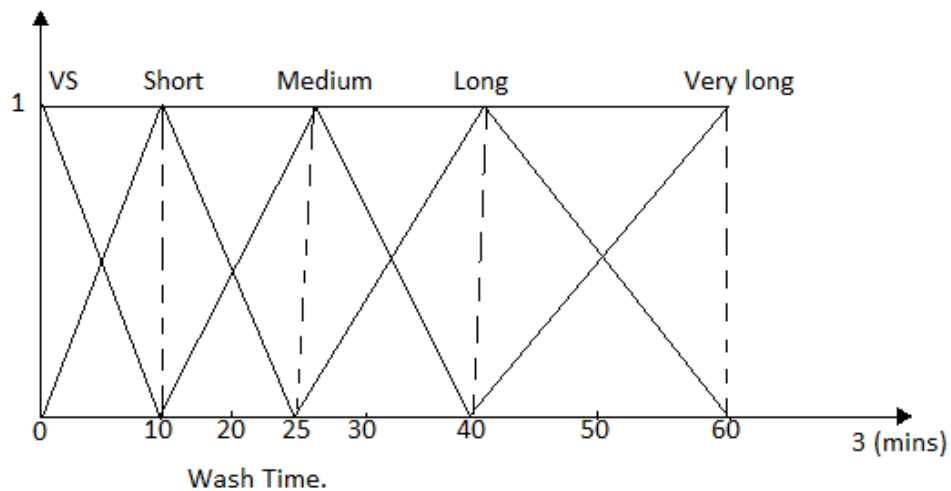


$$\mu_{\text{grease}}(y) = \mu_{\text{SG}}(y) = (50-y)/50, \quad 0 \leq y \leq 50$$

$$\begin{aligned} &= \mu_{\text{MG}}(y) = (y-0)/50, \quad 0 \leq y \leq 50 \\ &= (100-y)/50, \quad 50 < y \leq 100 \end{aligned}$$

$$= \mu_{\text{LG}}(y) = (y-50)/50, \quad 50 \leq y \leq 100$$

FUZZY CONTROLLER



$$\mu_{WT}(Z) = \mu_{VS}(z) = (10-z)/10, \quad 0 \leq z \leq 10$$

$$= \mu_S(z) = (z-0)/10, \quad 0 \leq z \leq 10$$

$$= (25-z)/(25-10), \quad 10 < z \leq 25$$

$$= \mu_M(z) = (z-10)/15, \quad 10 \leq z \leq 25$$

$$= (40-z)/15, \quad 25 < z \leq 40$$

$$= \mu_L(z) = (z-25)/15, \quad 25 \leq z \leq 40$$

$$= (60-z)/20, \quad 40 < z \leq 60$$

$$= \mu_{VL}(z) = (z-40)/20, \quad 40 \leq z \leq 60$$

FUZZY CONTROLLER

The sets of rules used to derive the output are:

Decide a fuzzy operator (AND or OR)

- a) If dirt is Large and grease is Large then wash_time is Very Long;
- b) If dirt is Medium and grease is Large then wash_time is Long;
- c) If dirt is Small and grease is Large then wash_time is Long;

FUZZY CONTROLLER

Step 3: Build a rule base

		y		
x		SG	MG	LG
SD		VS	M	L
MD		S	M	L
LD		M	L	VL

FUZZY CONTROLLER

Rule evaluation

Assume D= 60%, G =70%

$$\mu_{MD}(60) = (100-60)/50 = 40/50 = 4/5$$

$$\mu_{LD}(60) = (60-50)/50 = 10/50 = 1/5$$

$$\mu_{MG}(70) = (100-70)/50 = 30/50 = 3/5$$

$$\mu_{LG}(70) = (70-50)/50 = 20/50 = 2/5$$

Substitute the values in the rule decision table

$$\mu_{dirt}(x) = \mu_{SD}(x) = (50-x)/50, \quad 0 \leq x \leq 50$$

$$\begin{aligned} &= \mu_{MD}(x) = (x-0)/50, & 0 \leq x \leq 50 \\ &= (100-x)/50, & 50 < x \leq 100 \end{aligned}$$

$$= \mu_{LD}(x) = (x-50)/50, \quad 50 \leq x \leq 100$$

FUZZY CONTROLLER

Rule decision table

y				
x		SG	MG	LG
SD		x	x	x
MD		x	$\mu_M(z)$	$\mu_L(z)$
LD		x	$\mu_L(z)$	$\mu_{VL}(z)$

DEFUZZIFICATION

Fuzzification -> precise quantity to a fuzzy quantity

Defuzzification -> fuzzy quantity to a precise quantity

FUZZY CONTROLLER

Defuzzification (Max membership method)

1. Fuzzy operator, i.e. AND(min)
2. Take max value

$$\mu_{MD}(60) \cap \mu_{MG}(70) = \min(4/5, 3/5) = 3/5$$

$$\mu_{MD}(60) \cap \mu_{LG}(70) = \min(4/5, 2/5) = 2/5$$

$$\mu_{LD}(60) \cap \mu_{MG}(70) = \min(1/5, 3/5) = 1/5$$

$$\mu_{LD}(60) \cap \mu_{LG}(70) = \min(1/5, 2/5) = 1/5$$

$$\text{Max}(3/5, 2/5, 1/5, 1/5) = 3/5$$

y				
x		SG	MG	LG
SD	x	x	x	x
MD	x		$\mu_M(z)$	$\mu_L(z)$
LD	x		$\mu_L(z)$	$\mu_{VL}(z)$

FUZZY CONTROLLER

Step 4: Generate a crisp controlled output

$$\begin{aligned}\mu_M(z) &= (z-10)/15, & 10 \leq z \leq 25 \\ &= (40-z)/15, & 25 < z \leq 40\end{aligned}$$

$$(z-10)/15 = 3/5$$

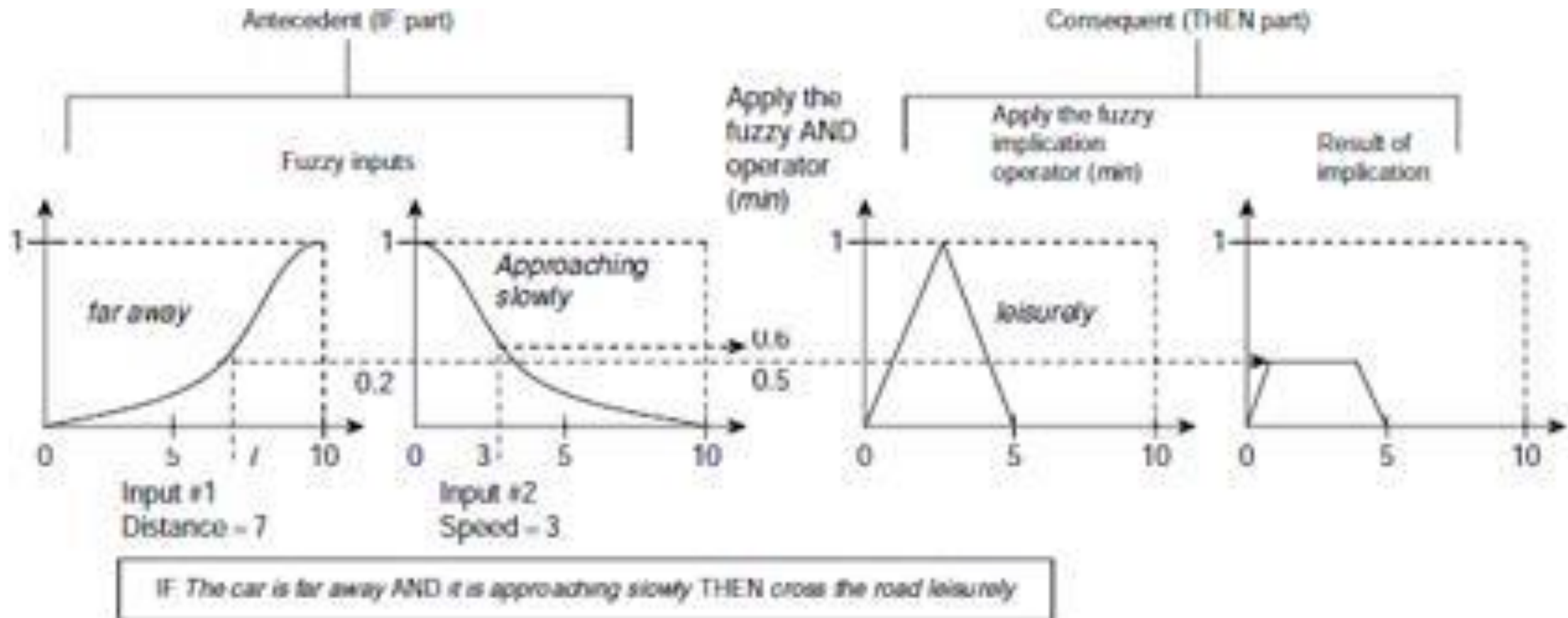
$$Z = 19$$

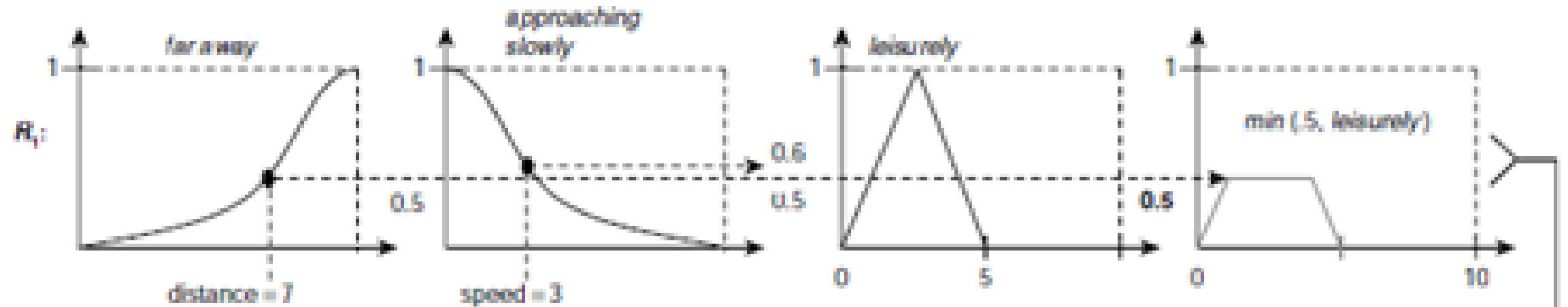
$$(40-z)/15 = 3/5$$

$$Z = 31$$

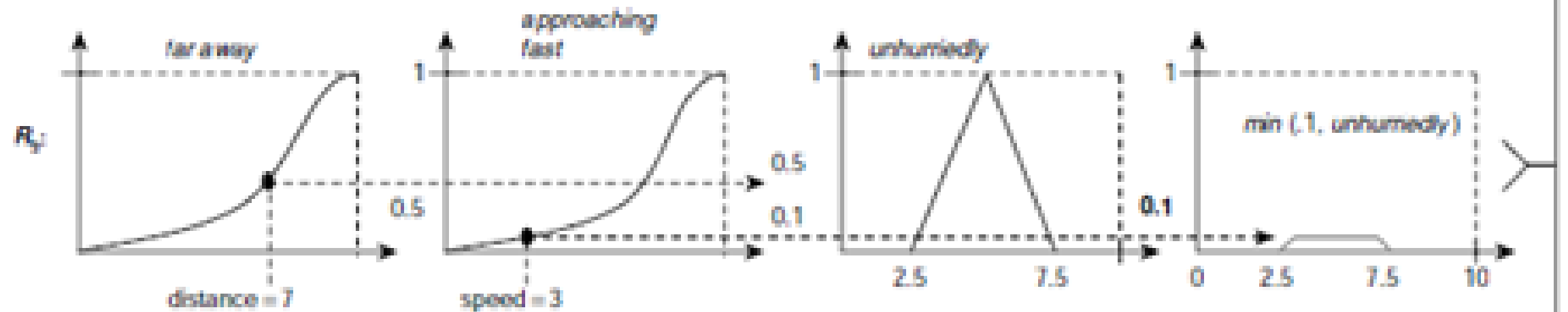
Therefore, $Z = (19+31)/2 = 25$ min

DEFUZZIFICATION METHODS



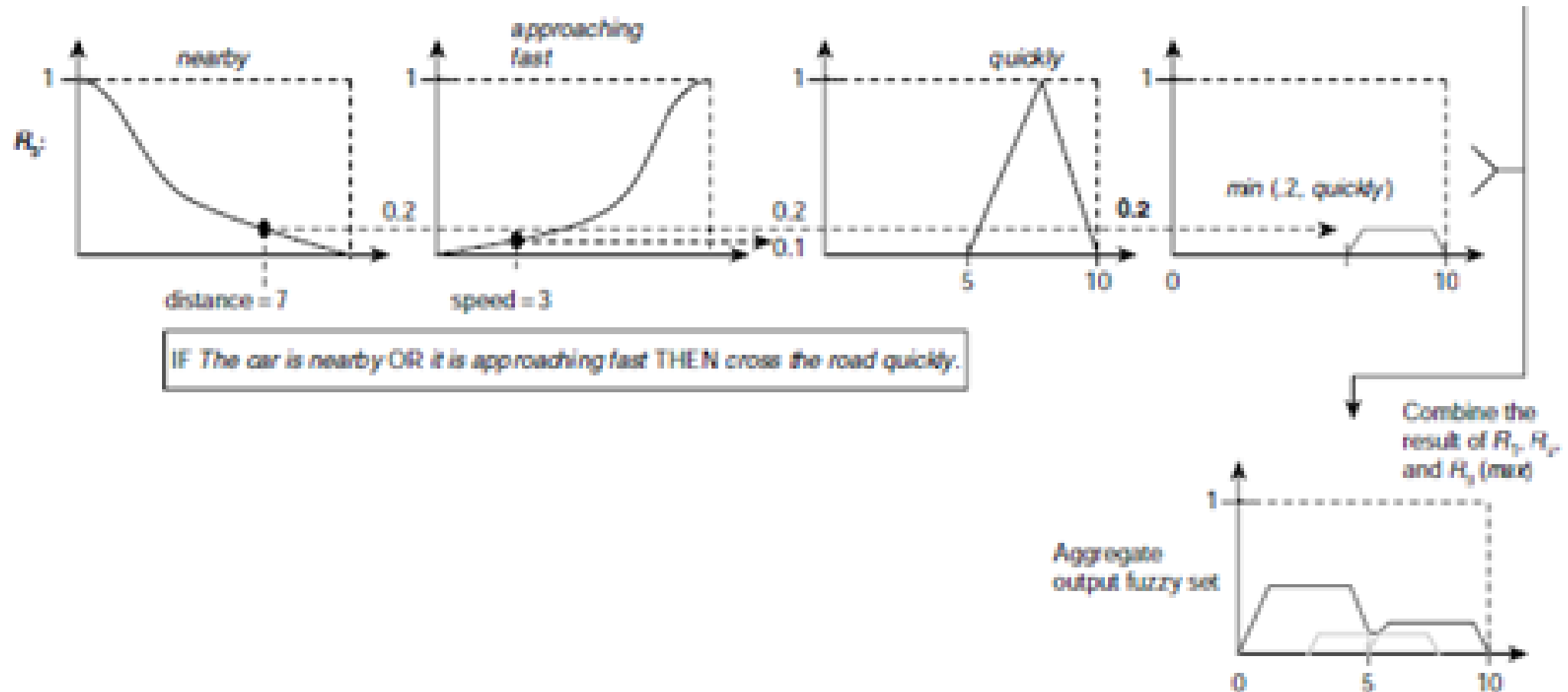


IF The car is far away AND it is approaching slowly THEN cross the road leisurely.

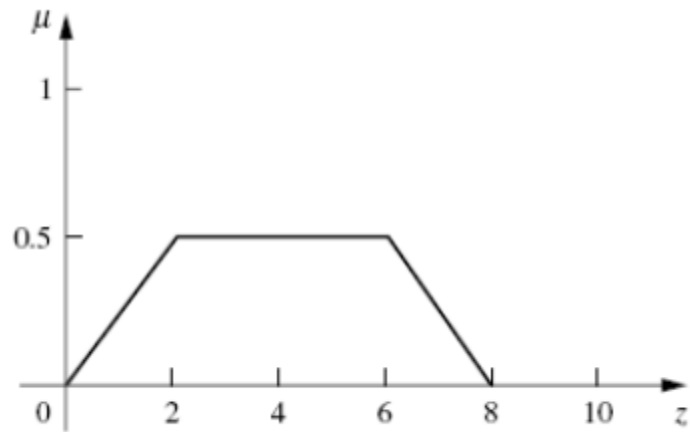


IF The car is far away AND it is approaching fast THEN cross the road unhurriedly.

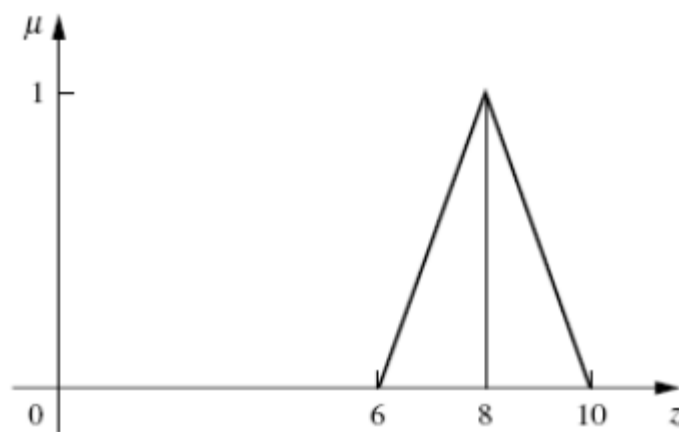
DEFUZZIFICATION METHODS



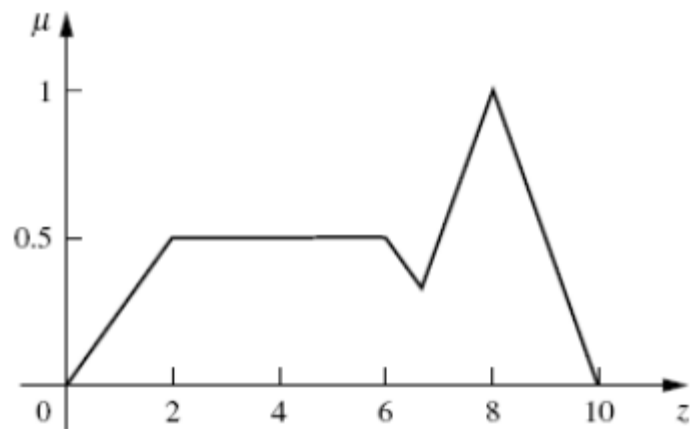
DEFUZZIFICATION METHODS



(a)



(b)

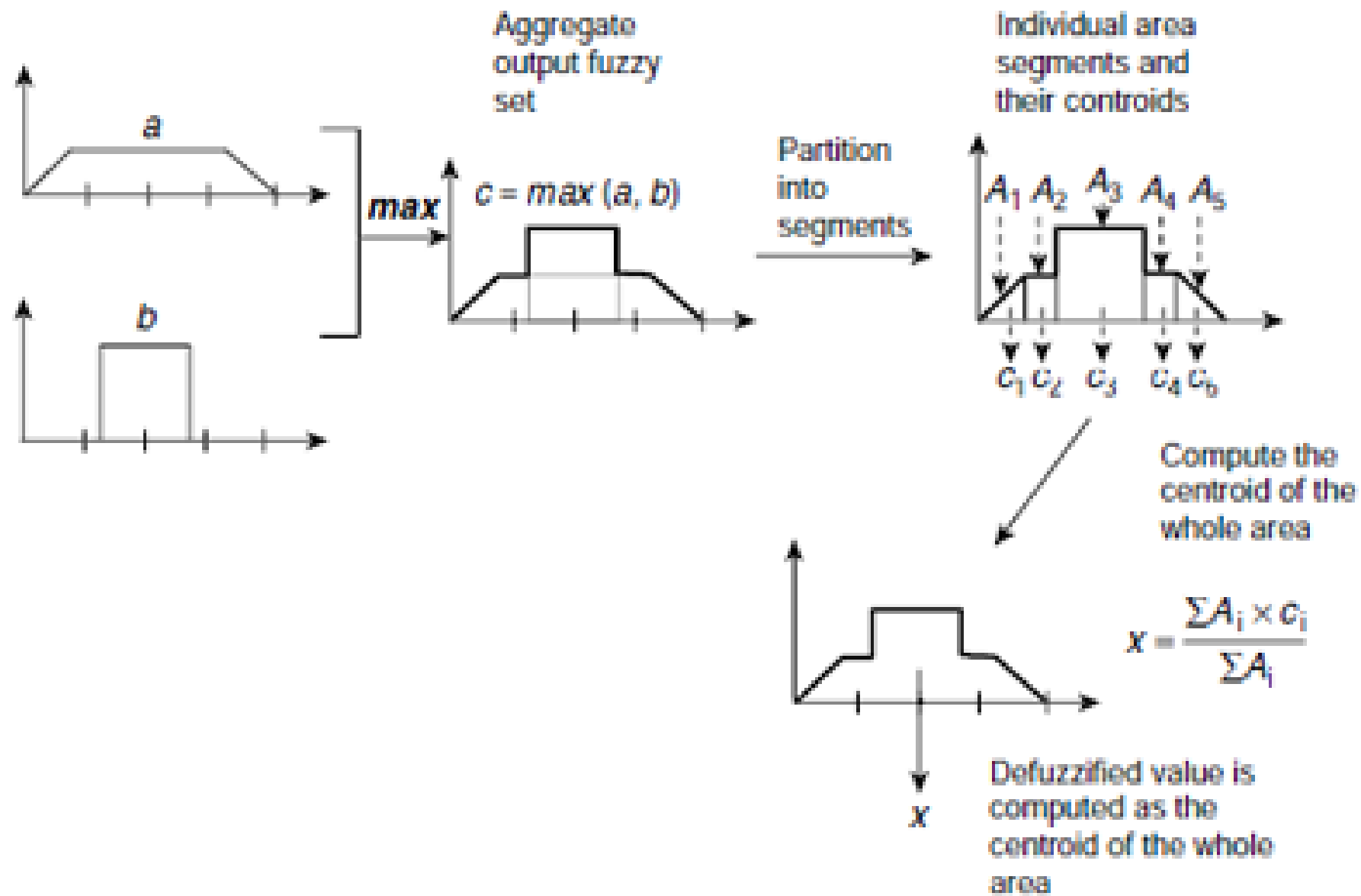


(c)

DEFUZZIFICATION METHODS

Centroid Method

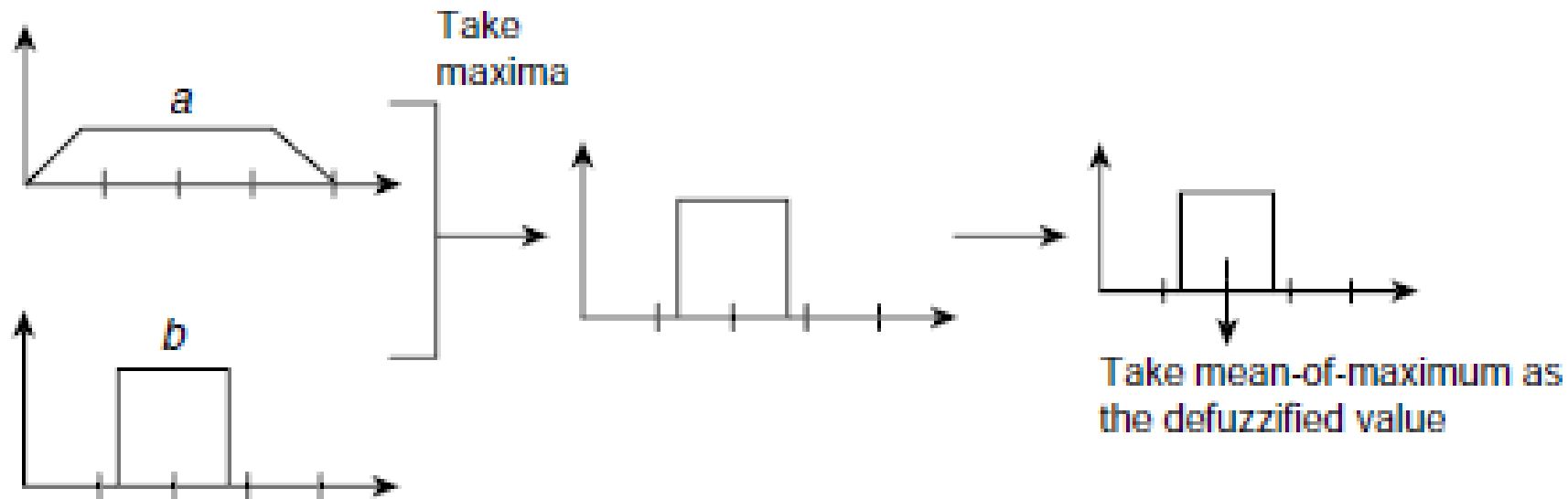
- The centroid method is also referred to as centre-of-gravity or centre-of-area method.
- The centroid, or centre-of-gravity, or centre-of-area of the resultant membership profile is taken as the defuzzified output



DEFUZZIFICATION METHODS

Mean-of-Maxima (MoM) Method

The highest degree of membership among all fuzzy sets is taken as the output defuzzified value.



DEFUZZIFICATION METHODS

Centre-of-Sums (CoS) Method

- The CoS method works almost in the same way as that of the centroid method
- This method employs the algebraic sum of the individual fuzzy subsets instead of their union.

